

Abstract

Projected freshening of Puget sound could influence the physiological performance in estuarine organisms. Differences in salinity tolerance between the native hairy shore crab (*Hemigrapsus oregonensis*) and the European green crab (*Carcinus maenas*) may affect their interactions as the environment continues to change. Previous studies have shown that the European green crab can tolerate a wide range of salinities, but comparative physiological responses remain poorly understood. This study examines hemolymph osmolality, righting time, and metabolic activity across three salinity treatments. Physiological responses were found to be more driven by species identity rather than by salinity, suggesting that species specific physiological traits are more pronounced than salinity effects under the conditions tested.

Introduction

Puget Sound is a large estuarine ecosystem that is increasingly threatened by climate-driven environmental change and the expansion of the European green crab, an invasive species (Fisher et al., 2024; Walker et al., 2022). Climate models predict that rising temperatures will lead to a freshening of Puget Sound (Walker et al., 2022), which may place physiological stress on the native hairy shore crab while simultaneously favoring the invasive European green crab. Together, these stressors may alter trophic interactions and ecosystem stability.

The projected freshening of Puget Sound raises concerns for the hairy shore crab because previous studies indicate physiological stress in low salinity conditions. Past studies have shown increased oxygen consumption in hairy shore crabs kept in 25 ppt seawater, regardless of the temperature (DEHNEL, 1960). In contrast, European green crabs have exhibited a wide range of salinity tolerance and are known to inhabit waters between 1.4 - 54 ppt (Leignel et al., 2014). These differences in salinity tolerance are of ecological concern because in Washington the hairy shore crab is a common prey item for the European green crab (Fisher et al., 2024), meaning that future freshening may provide the invasive European green crab with a competitive and predatory advantage over the native hairy shore crab.

In this study, we examine the physiological responses to salinity changes in both the hairy shore crab and the European green crab in order to evaluate how the native *H. oregonensis* may respond to the projected freshening of Puget Sound and to assess whether if *C.maenas* could potentially outcompete the hairy shore crab. We do so by measuring the righting time, osmolality, and metabolic activity for both species of crabs.

Methods

Experimental overview

A total of 15 crabs consisting of 12 hairy shore crabs and 3 European green crabs were used for this study. Crabs were assigned to one of three salinity treatments consisting of 15 ppt, 35 ppt, or 45 ppt seawater for a period of three weeks. Each salinity treatment had one European

green crab and four hairy shore crabs. Crab weight, righting time, and metabolic activity was measured weekly. Hemolymph extractions were performed on hairy shore crabs during week one to determine a baseline and on all crabs during week three to compare.

Righting time and weight

After a crab was weighed it was placed dorsal side down on a flat surface and the time required to right itself was recorded.

Hemolymph extractions

Hemolymph was extracted from crabs by inserting a small needle into the joint of a crab leg. The extracted hemolymph was then analyzed using an osmometer.

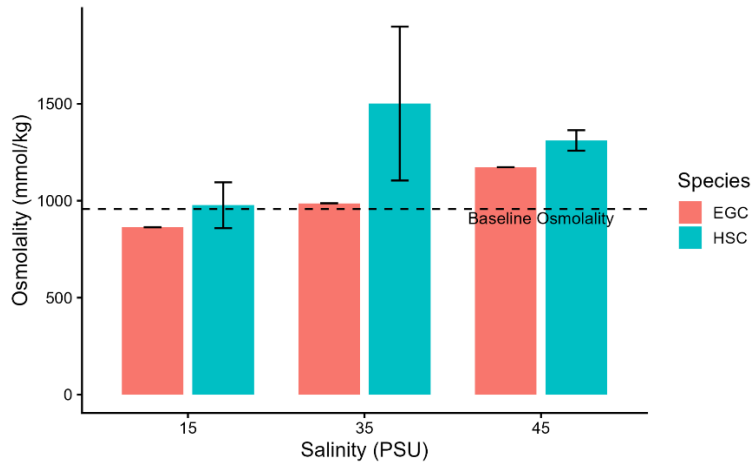
Respiration rates

Metabolic activity was measured using a resazurin assay. Individual crabs were placed into separate chambers with enough resazurin to submerge them completely, along with a blank chamber containing no crab for each species. Aliquots of 200 ul were collected at 0, 30, and 60 minute time intervals and transferred to a 96 well plate. Fluorescence was measured using a spectrophotometer plate to quantify resorufin production. Afterwards, the crabs were rinsed twice and placed back into their respective salinity conditions.

Results

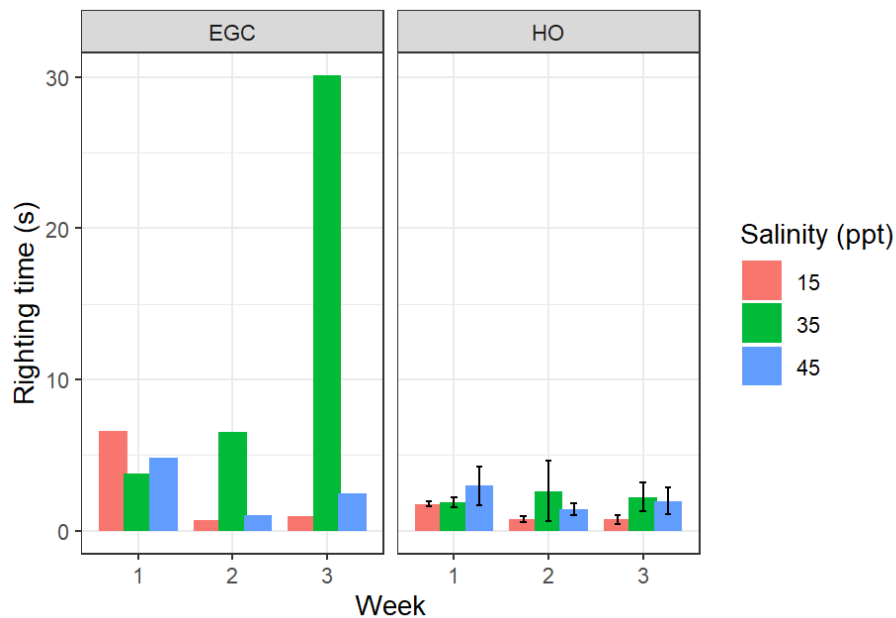
To compare physiological responses to salinity changes between *C. maenas* and *H. oregonensis*, a total of 15 crabs were used. This includes 12 *H. oregonensis* and 3 *C. maenas*. A two-way ANOVA was performed to assess differences in hemolymph osmolality among the salinity treatment groups (fig.1). No statistical differences were observed between any species-salinity pairs. However, two-sided t-tests comparing treatment groups to the average baseline osmolality showed that *H. oregonensis* crabs in 45 PPT treatment had a significantly higher osmolality (p-value = 0.006785, df = 3). Hairy shore crabs in other salinity treatments were not significantly different from the baseline.

Figure 1. Hemolymph osmolality in *H. oregonensis* and *C. maenas*



Osmolality (mmol/kg) of hemolymph after three weeks in sustained salinity treatments (15, 35, 45 ppt). Red bars represent *C.maenas* (EGC) and blue bars represent *H. oregonensis*, standard error bars are used where applicable.

A three-way ANOVA identified significant differences in righting time between species, salinity and week (Table1; Fig 2) . A Tukey post hoc test revealed that the European green crab kept in 35 ppt conditions during week 3 showed a significantly different righting time when compared to other treatment groups. A two-way ANOVA uncovered significant differences between species and metabolic activity ($p\text{-value} = 8.14\text{e-}07$, $df = 1$). However, neither salinity treatment or the interaction between species and salinity had a significant effect on metabolic activity (Fig 3). **Figure 2: Righting Time of *H. oregonensis* and *C. maenas***

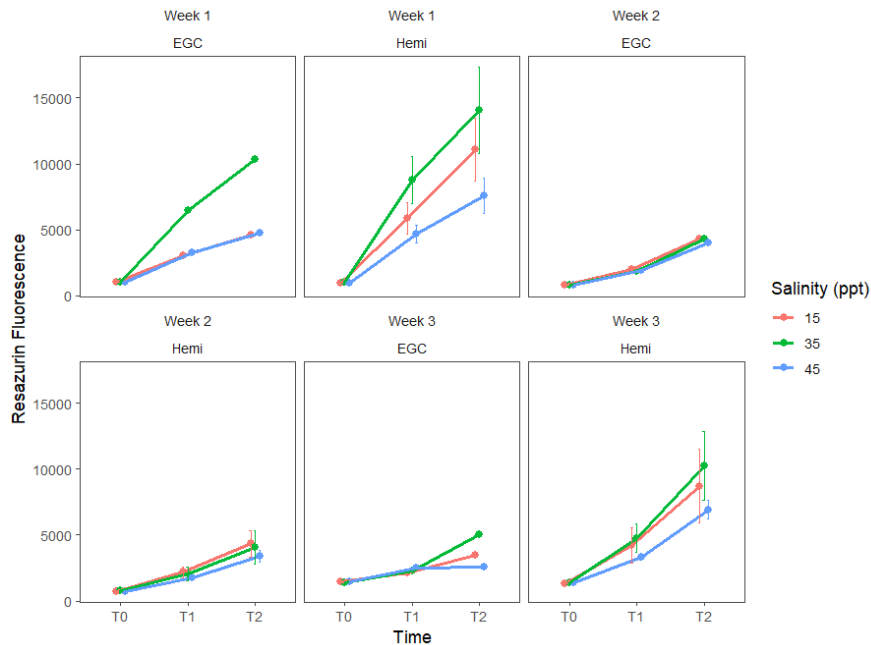


Righting time (s) of *H. oregonensis* (HO) and *C.maenas* (EGC) under varying salinity conditions across three weeks (15, 35, 45 ppt) Bars represent the righting time for each species and treatment combination, with standard error bars where applicable.

Table 1. Righting time ANOVA summary

Effect	df	F value	p-value
Species	1	55.31	1.12e-07
Salinity	2	14.56	7.21e-05
week	2	4.828	0.017
Species x salinity	2	31.01	2.23e-07
Salinity x week	4	9.25	1.14e-4
Species x week	2	17.41	2.13e-05
Species x salinity x week	4	23.38	5.55e-08

Figure 3. Resorufin Production in *H. oregonensis* and *C. maenas* Over Time



Resorufin production for each salinity treatment in *H. oregonensis* (HO) and *C. maenas* (EGC) measured after being submerged for 30-minute intervals.

The results presented here describe physiological responses to salinity variation in *H. oregonensis* and *C. maenas*. Osmolality showed a single significant increase in *H. oregonensis* at 45 ppt relative to the baseline, righting time differed across species, salinity, and time with multiple significant pairwise differences, and metabolic activity varied significantly between species but not across salinity treatments.

Discussion

This study examined physiological responses to salinity changes in *H. oregonensis* and *C. maenas* using hemolymph osmolality, righting time, and metabolic activity. These comparisons were used to evaluate differences in species performance under varying salinity conditions and how they may influence competitive interactions between invasive and native species. Hemolymph osmolality had limited variation overall, with only *H. oregonensis* at 45 ppt showing a significant increase relative to the baseline, which may simply reflect the species' ability to adjust its hemolymph solute concentration in response to changing salinity conditions (DEHNEL, 1962). Righting time varied significantly across species, salinity and time, with the European green crab in 35 ppt treatment during week three showing a delayed response. Metabolic activity also differed significantly between the two species but was not affected by salinity. It's possible that molting in one European green crab contributed to variation in both righting time and metabolic activity between species.

Between weeks two and three, the European green crab in 35 ppt molted. This may have influenced metabolic activity during the week three resazurin assay, as molting in crustaceans disrupts energy homeostasis due to the increased energetic demands associated with post-molt recovery (Yuan et al., 2022). This also greatly affected righting time during week three, as the crab took 30 seconds to right itself. It's entirely possible that the significant difference between species and righting time is due to this molt, as past studies have shown that crabs are significantly weaker after molting (Taylor et al., 2007).

Further investigation into physiological responses across fluctuating salinity conditions and larger sample sizes may help clarify how *H. oregonensis* and *C. maenas* might respond to the projected freshening of Puget Sound. Overall, the results of this study indicate physiological performance was shaped by species rather than variations in salinity, which may have implications for competitive interactions between the hairy shore crab and the European green crab. However, the extent as to which these differences translate to long term ecological outcomes under changing environmental conditions remains uncertain.

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